

Performance modeling of CS

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Renewal Processes

A Renewal process is a counting process for which the inter-arrival times are i.i.d with an arbitrary distribution.

Renewal process is a generalization of Poisson process where the inter-arrival times were i.i.d exponential.

Renewal Processes - Notations

- ▶ $\{X_n, n \geq 0\}$ denote the sequence of inter-arrival times of a renewal process.
- ▶ X_n is the time between $n - 1$ th and n th renewal.
- ▶ $\{X_n, n \geq 0\}$ are non negative iid random variables with law F .
- ▶ Let $S_0 = 0$ and $S_n = \sum_{i=1}^n X_i$

The term renewal process refers to any of the following:

- 1) The sequence $\{X_n, n \geq 0\}$ of inter-arrival times
- 2) The sequence $\{S_n, n \geq 0\}$ of arrival times
- 3) The associated counting process $\{N(t), t \geq 0\}$.

Renewal Process Examples

- ▶ Bernoulli/Binomial process.
- ▶ Poisson process.
- ▶ Successive times between your water bottle gets empty.
- ▶ Time instants when number of customers in Ikea is exactly 10.
- ▶ Time between successive visits to a particular state of a Markov Chain.

Relation between S_n and $N(t)$

- ▶ Define $N(t) = \sup\{n : S_n \leq t\}$. $N(t)$ signifies the number of renewals until time t .

$$N(t) \geq n \Leftrightarrow S_n \leq t$$

- ▶ $P\{N(t) \geq n\} = P\{S_n \leq t\}$
- ▶ $P\{N(t) = n\} = P\{N(t) \geq n\} - P\{N(t) \geq n + 1\}.$
- ▶ $P\{N(t) = n\} = P\{S_n \leq t\} - P\{S_{n+1} \leq t\}.$
- ▶ $P\{N(t) = n\} = F_n(t) - F_{n+1}(t).$
- ▶ How do you obtain F_n from F ?

Convolution basics

- ▶ Convolution of two functions functions:

$$(f * g)(t) := \int_{-\infty}^{\infty} f(t - u)g(u)du$$

- ▶ Convolution of two positive functions:

$$(f * g)(t) := \int_0^t f(t - u)g(u)du.$$

- ▶ Convolution of a function w.r.t a distribution:

$$(f * G)(t) := \int_{-\infty}^{\infty} f(t - u)dG(u) = \int_{-\infty}^{\infty} f(t - u)g(u)du$$

Convolution basics

- ▶ Convolution of distributions is also a distribution.
- ▶ If F and G are two distributions then
$$(F * G)(t) := \int_{-\infty}^{\infty} F(t - u) dG(u). \quad F^{(*2)}(t) = (F * F)(t).$$
- ▶ $P(S_n \leq t) := F_n(t)$. We will now express $F_n(t)$ as a convolution!
- ▶ $P(S_1 \leq t) = P(X_1 \leq t) = F(t)$ where $F(t)$ is the cdf of the interarrival time X_1 .
- ▶ $P(S_2 \leq t) = P(X_1 + X_2 \leq t) = \int_0^t F(t - u) f(u) du$
- ▶ $= \int_0^t F(t - u) dF(u) = (F * F)(t).$

$$F_n(t) = F^{(*n)}(t)$$

Laplace transform basics

- ▶ Bilateral Laplace transform of function $f(\cdot)$ is given by $\bar{f}(s) := \int_{-\infty}^{\infty} e^{-st} f(t) dt$.
- ▶ Consider a random variable X with distribution F . Then recall that $M_X(s) = E[e^{sX}] = \int e^{st} dF(t)$.
- ▶ The laplace transform of a distribution F is defined as $\bar{F}(s) = \int_{-\infty}^{\infty} e^{-st} dF(t) = E[e^{-sX}]$. Here X is a random variable with distribution F .
- ▶ Property: Consider $Z(t) = (f * F)(t)$. Then $\bar{Z}(s) = \bar{f}(s)\bar{F}(s)$
- ▶ This implies that if $Z(t) = (F * F)(t)$, then $\bar{Z}(s) = \bar{F}^2(s)$.
- ▶ Then by the same logic, $LT\{F^{(*n)}(t)\} = \bar{F}^n(s)$.

Renewal equation $m(t)$

Let $m(t)$ denote the mean number of arrivals by time t , i.e., $m(t) := E[N(t)]$. Then $m(t) = \sum_{n=1}^{\infty} F_n(t)$.

What is $m(t)$ for the Poisson process?

Let $\bar{m}(s)$, $\bar{F}(s)$ and $\bar{F}_n(s)$ denote the Laplace transform of $m(t)$, $F(t)$ and $F_n(t)$ respectively. Then $\bar{m}(s) = \frac{\bar{F}(s)}{1 - \bar{F}(s)}$.

- ▶ $\bar{m}(s) = \bar{F}(s) + \bar{m}(s)\bar{F}(s)$. Inverse Laplace transform gives
- ▶ $m(t) = F(t) + (m * F)(t)$.

Renewal equation

- ▶ Renewal equation is an integral equation for $m(t)$ that is obtained by conditioning on time for first renewal.
- ▶ Suppose $X_1 = x$. Since this is the time interval between 0th and 1st arrival, $S_1 = x$ and the first arrival has happened at x .
- ▶ $m(t) = E[N(t)] = E_F[E[N(t)/X_1]]$.
- ▶ Therefore $m(t) = \int_0^\infty E[N(t)/X_1 = x]dF(x)$
- ▶ What if $t < x$? Then $E[N(t)/X_1 = x] = 0$.
- ▶ What happens when $t \geq x$?
- ▶ $E[N(t)/X_1 = x] = 1 + m(t - x)$.
- ▶ This gives us the renewal equation
$$m(t) = \int_0^t (1 + m(t - x))dF(x).$$